26900353 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

Regular EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEMATA00 - SUPERVISED MACHINE LEARNING: ALGORITHMS AND APPLICATIONS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One or Two Sentences

Answer any **5** questions

Each question carries **2** marks

1. How does Naive Bayes determine the most likely class for a new data point? [U] / [CO1]
2. What happens if the value of K is chosen to be very small in KNN? [U] / [CO1]
3. Which feature will be selected for splitting? [U] / [CO2]
4. What is meant by persisting a decision tree? [U] / [CO2]
5. What is the main objective of SVM in a binary classification problem? [U] / [CO3]
6. What is Sequential Minimal Optimization (SMO) in SVM? [U] / [CO3]
7. Differentiate between BaggingClassifier and BaggingRegressor. [An] / [CO4]
8. Name any two hyperparameters commonly tuned in AdaBoost. [U] / [CO4]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **6** marks

9. Write a Python program using KNeighborsClassifier with K=3 to classify the new data point (4,4) using the following dataset. Display the predicted class. [A] / [CO1]

X_1	X_2	Class
1	1	A
2	2	A
5	5	B
6	6	B

10. Explain the role of distance measures in KNN classification. Discuss any two commonly used distance measures with their mathematical expressions. [U] / [CO1]

11. Explain the working of the `plotNode()` and `createPlot()` functions used in the decision tree plotting program. [A] / [CO2]

12. Explain the concept of a Decision Tree,using the email classification example. [K] / [CO2]

13. Explain how support vectors influence the position of the optimal separating hyperplane. What would happen to the decision boundary if a non-support vector were removed from the training dataset? [U] / [CO3]

14. Explain the basic idea of the kernel trick in SVM. How does it help in handling non-linearly separable data? [U] / [CO3]

15. Explain the Letter Recognition dataset used for Random Forest classification. [An] / [CO4]

16. Discuss the applications of the Extra Trees algorithm. [U] / [CO4]

[6x5 = 30]

Part C

Essay Questions

Answer any **3** questions

Each question carries **10** marks

17. Consider the following dataset. Write a Python program using GaussianNB to divide the data into training and testing sets, train the classifier, predict the classes of the test data, and evaluate the model using accuracy and a confusion matrix. [A] / [CO1]

X_1	X_2	Class
1	1	A
2	2	A
2	3	A
3	2	A
6	6	B
7	5	B
7	7	B
8	6	B

18. Explain the concept of persisting a decision tree. Describe the `storeTree()` and `grabTree()` functions and discuss their importance. [An] / [CO2]

19. Explain the `plotTree()` and `createPlot()` functions in detail. Describe how they work together to draw a complete decision tree. [An] / [CO2]

20. Consider the following linearly separable dataset. Write a Python program to plot the data points and determine a suitable separating hyperplane. Explain how the maximum margin hyperplane can be identified and indicate the support vectors. [A] / [CO3]

x_1	x_2	Class
1	1	A
2	2	A
2	3	A
4	5	B
5	6	B
6	5	B

21. What is the error term in Platt's SMO algorithm? Explain how it is used during the optimization process. [U] / [CO3]

22. Explain the **Bagging (Bootstrap Aggregating)** algorithm in detail. Discuss the following: [U] / [CO4]

    - Ensemble learning
    - Bootstrap sampling
    - Training multiple Decision Trees
    - Majority voting for classification
    - Advantages and limitations of Bagging
    - Applications of Bagging

[10x3 = 30]